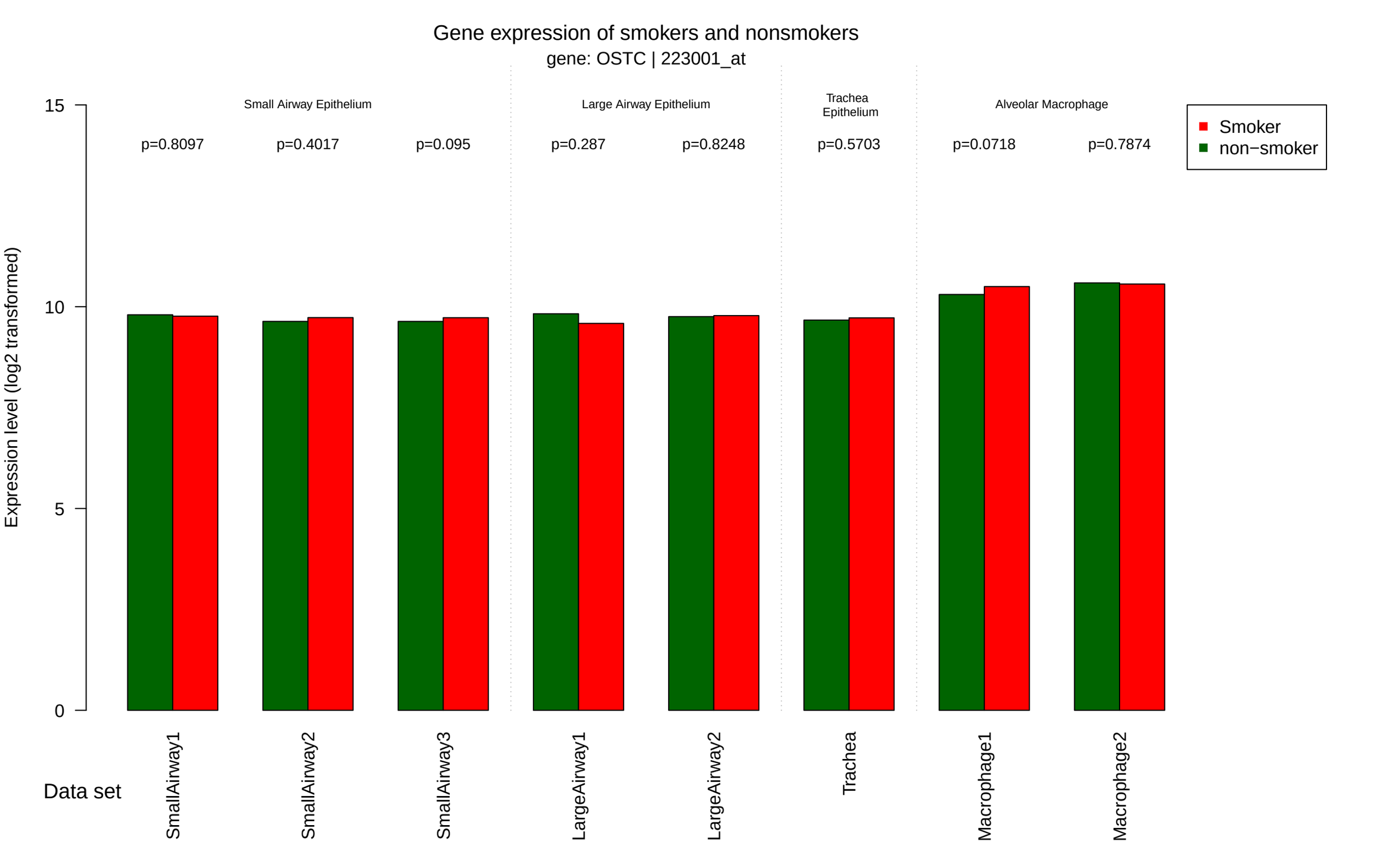
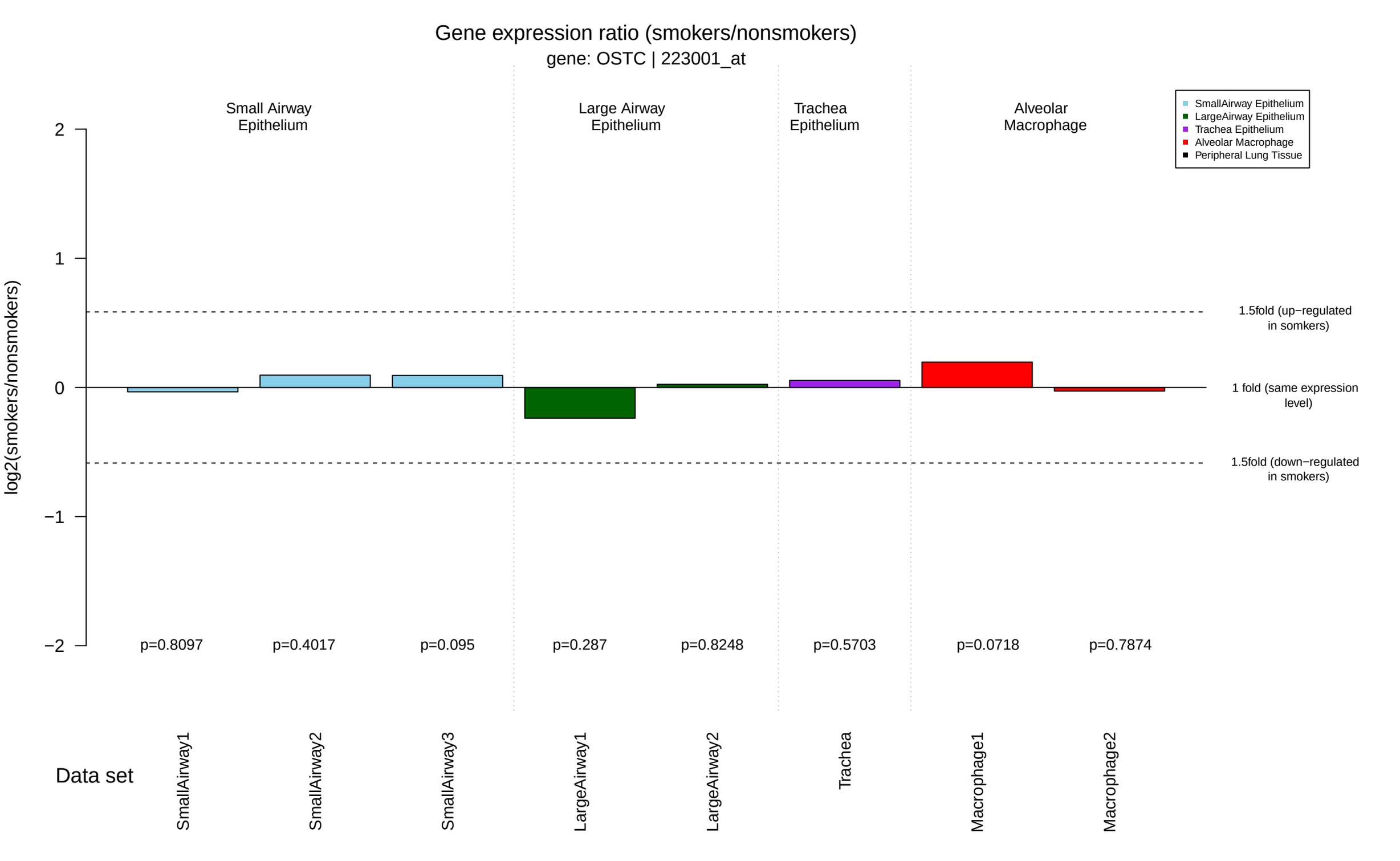
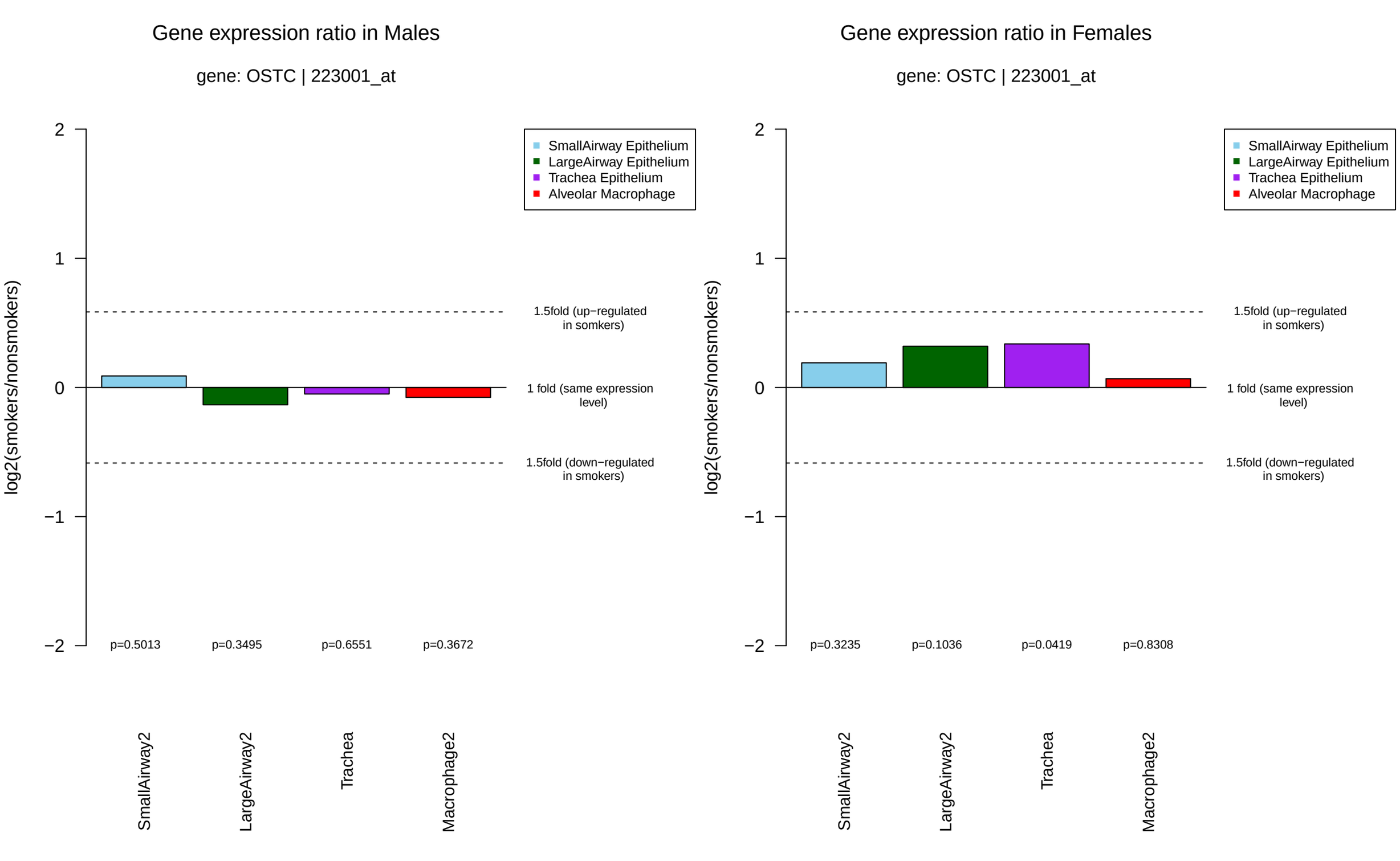
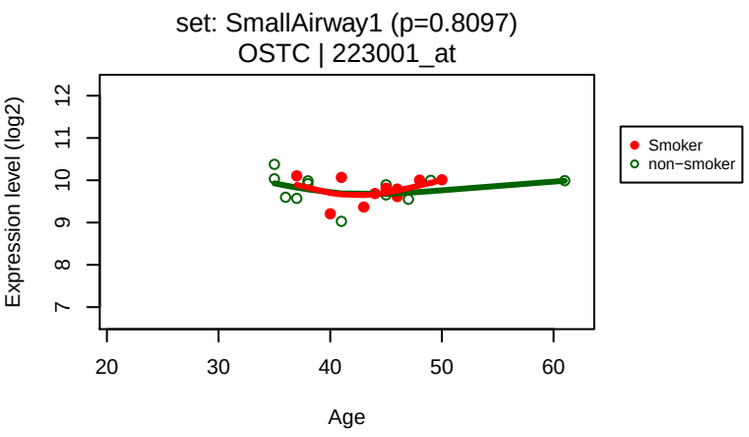
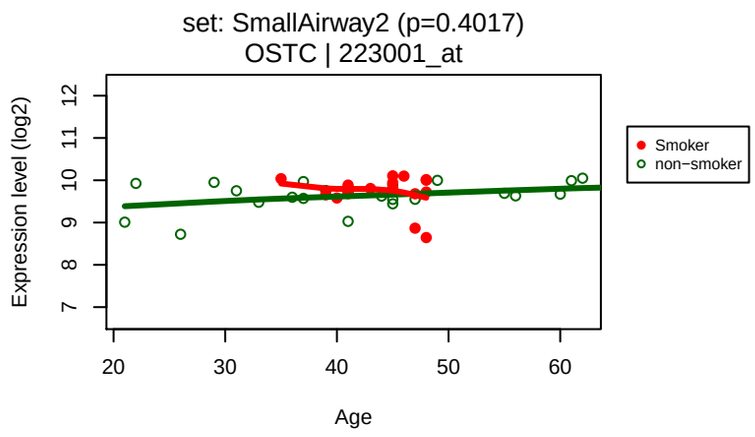
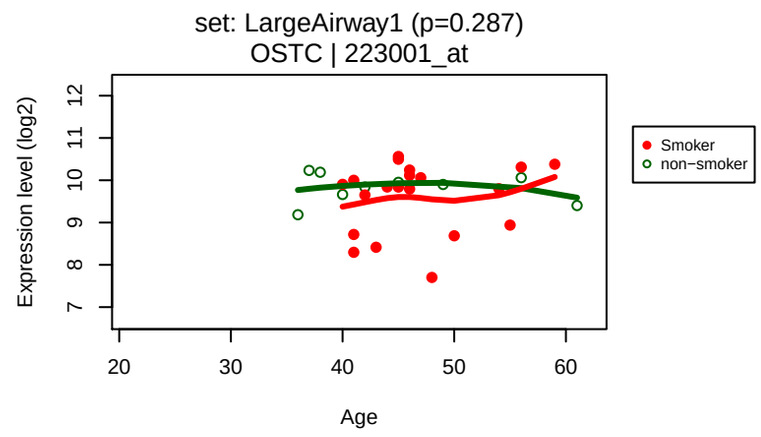

Gene expression of smokers and nonsmokers
gene: OSTC | 223001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8097
p=0.4017
p=0.095
p=0.287
p=0.8248
p=0.5703
p=0.0718
p=0.7874
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

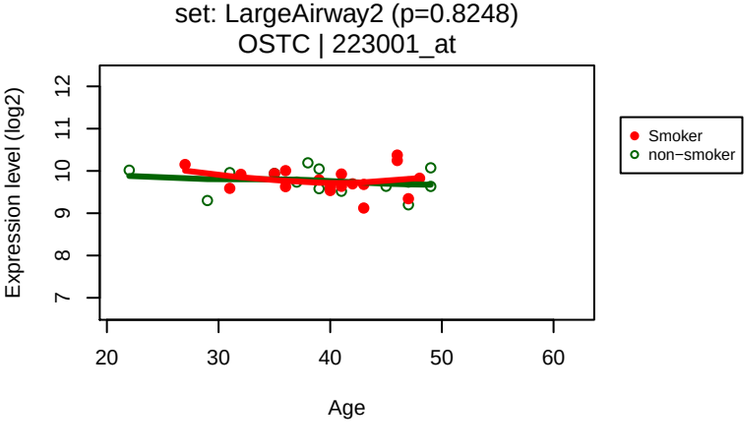

Gene expression ratio (smokers/nonsmokers)
gene: OSTC | 223001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8097
p=0.4017
p=0.095
p=0.287
p=0.8248
p=0.5703
p=0.0718
p=0.7874
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

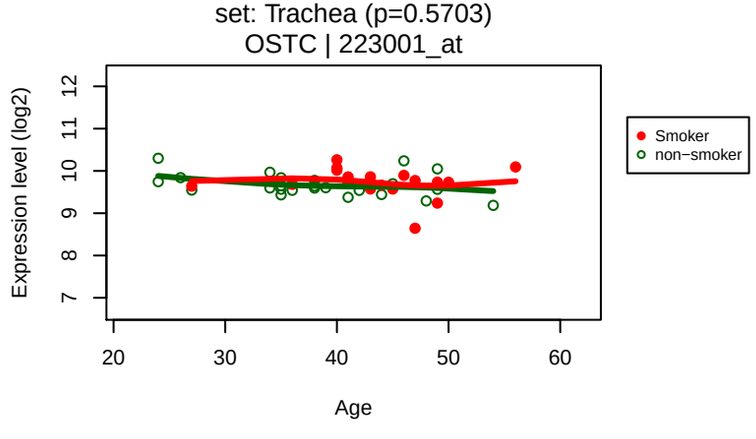

Gene expression ratio in Males
gene: OSTC | 223001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5013
p=0.3495
p=0.6551
p=0.3672
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSTC | 223001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3235
p=0.1036
p=0.0419
p=0.8308
SmallAirway2
LargeAirway2
Trachea
Macrophage2

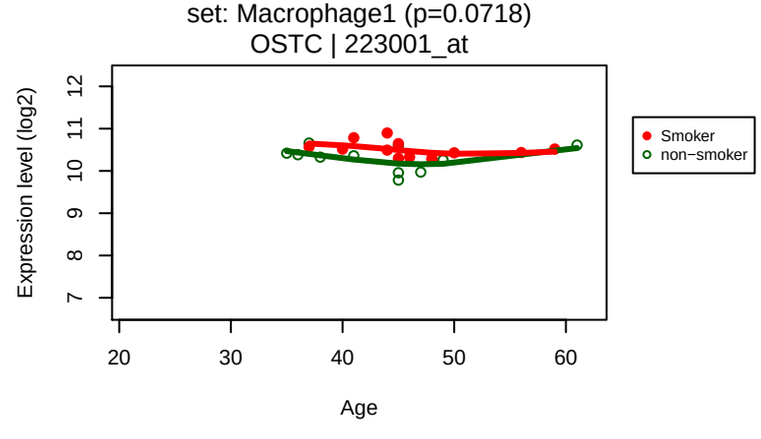

set: SmallAirway1 (p=0.8097)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker

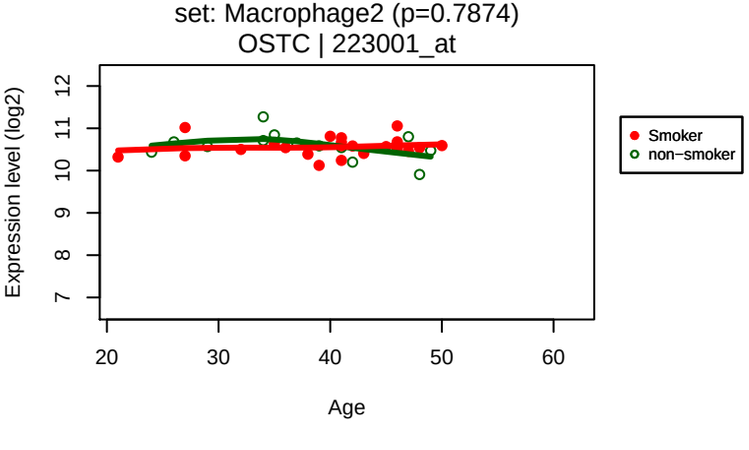

set: SmallAirway2 (p=0.4017)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker

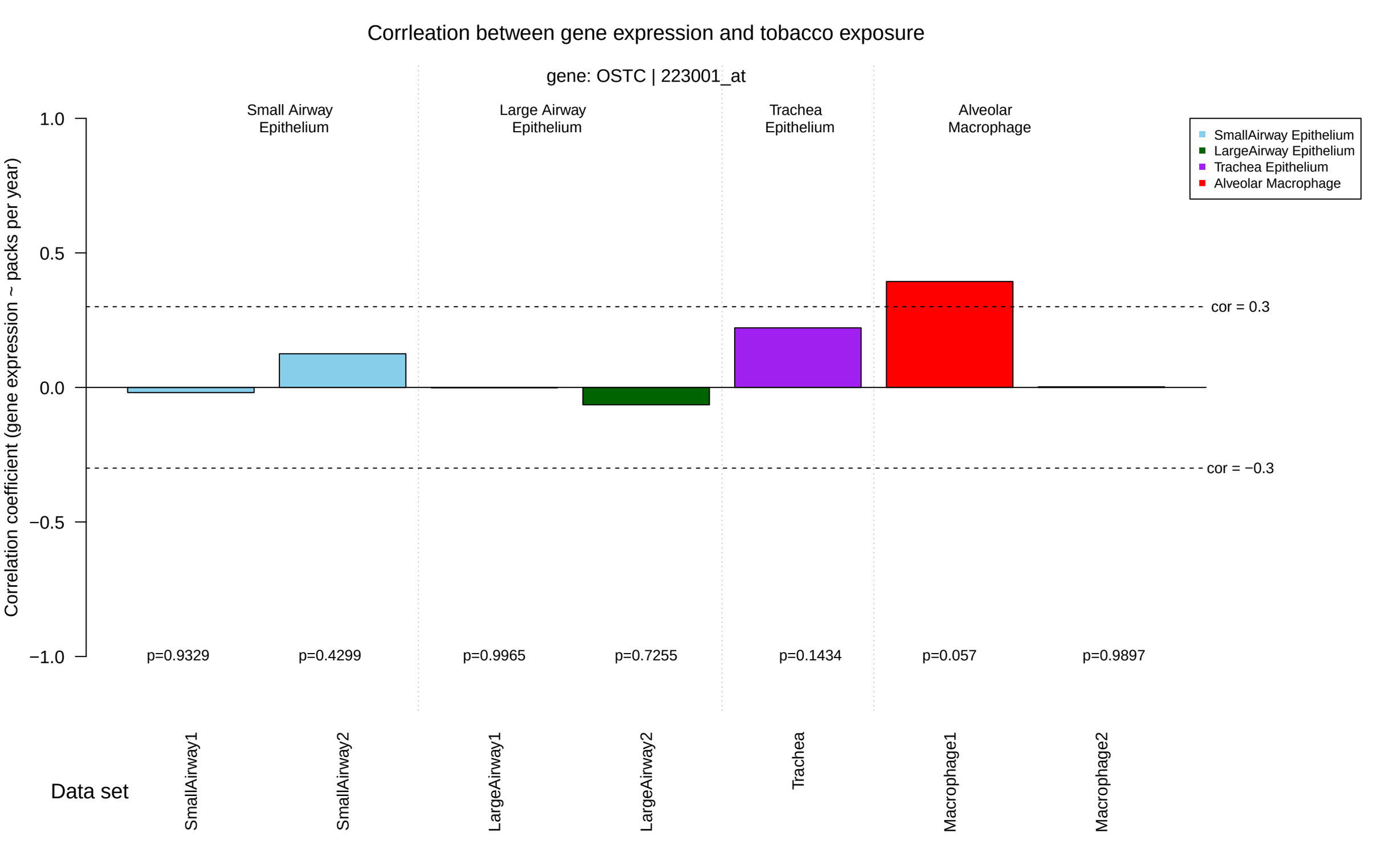

set: LargeAirway1 (p=0.287)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8248)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5703)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0718)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7874)
OSTC | 223001_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSTC | 223001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9329
p=0.4299
p=0.9965
p=0.7255
p=0.1434
p=0.057
p=0.9897
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2